Date: 23/07/2026

Lab Practical #07:

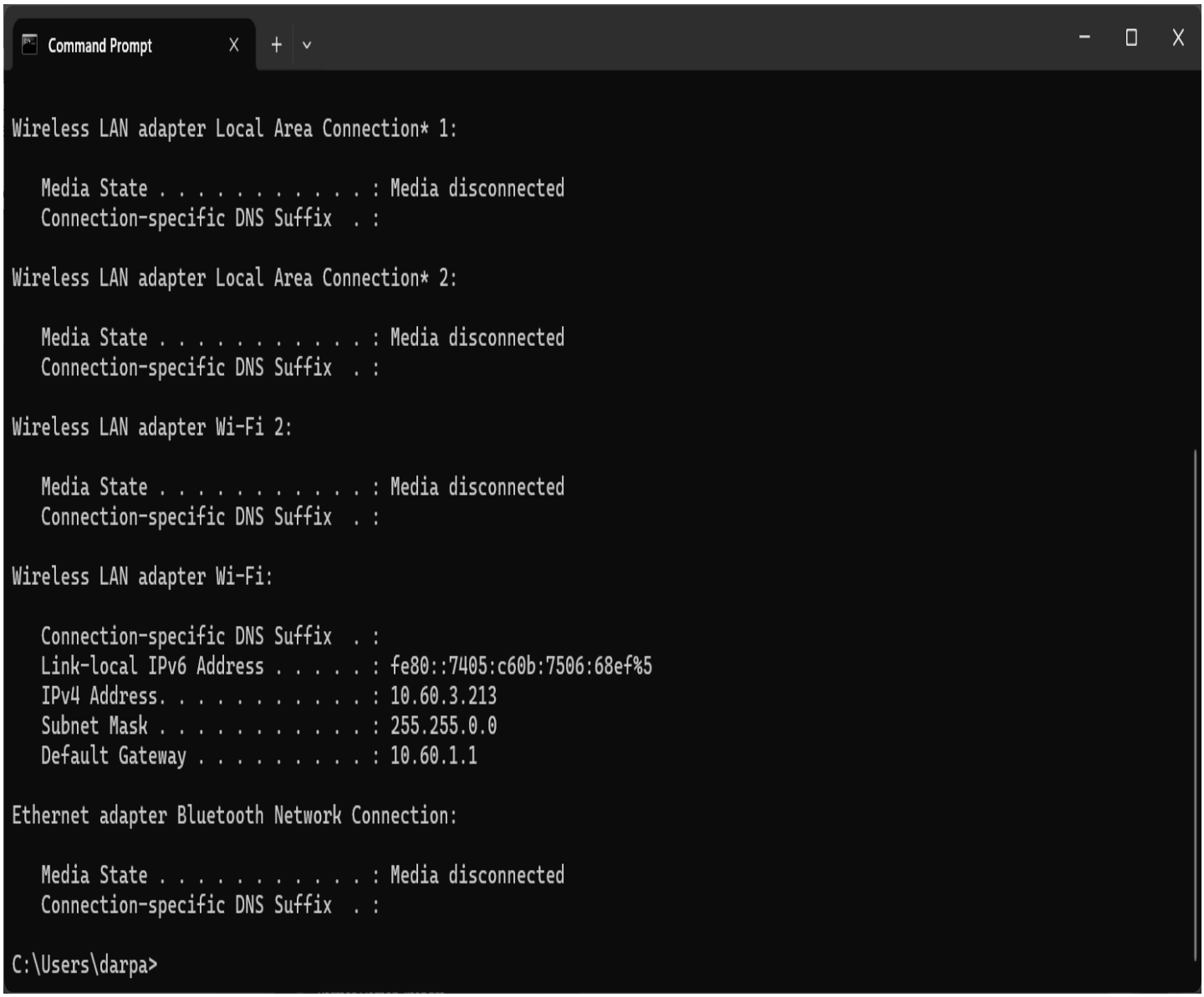
To configure Telnet & SSH for remote connection.

Practical Assignment #07:

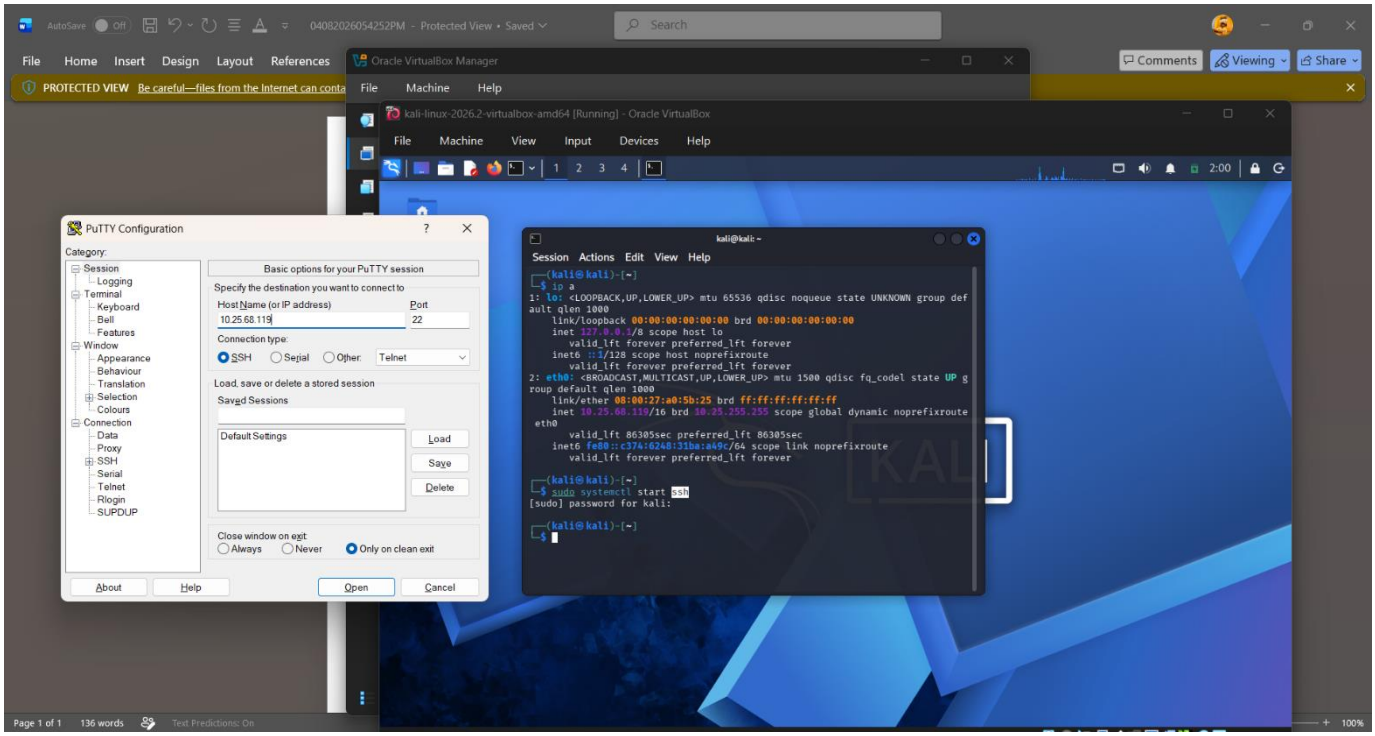
1. **To configure Telnet and SSH services on a network topology, perform remote login operations, and capture network packets to analyze security differences (plain text vs. encryption).**

Instructions:

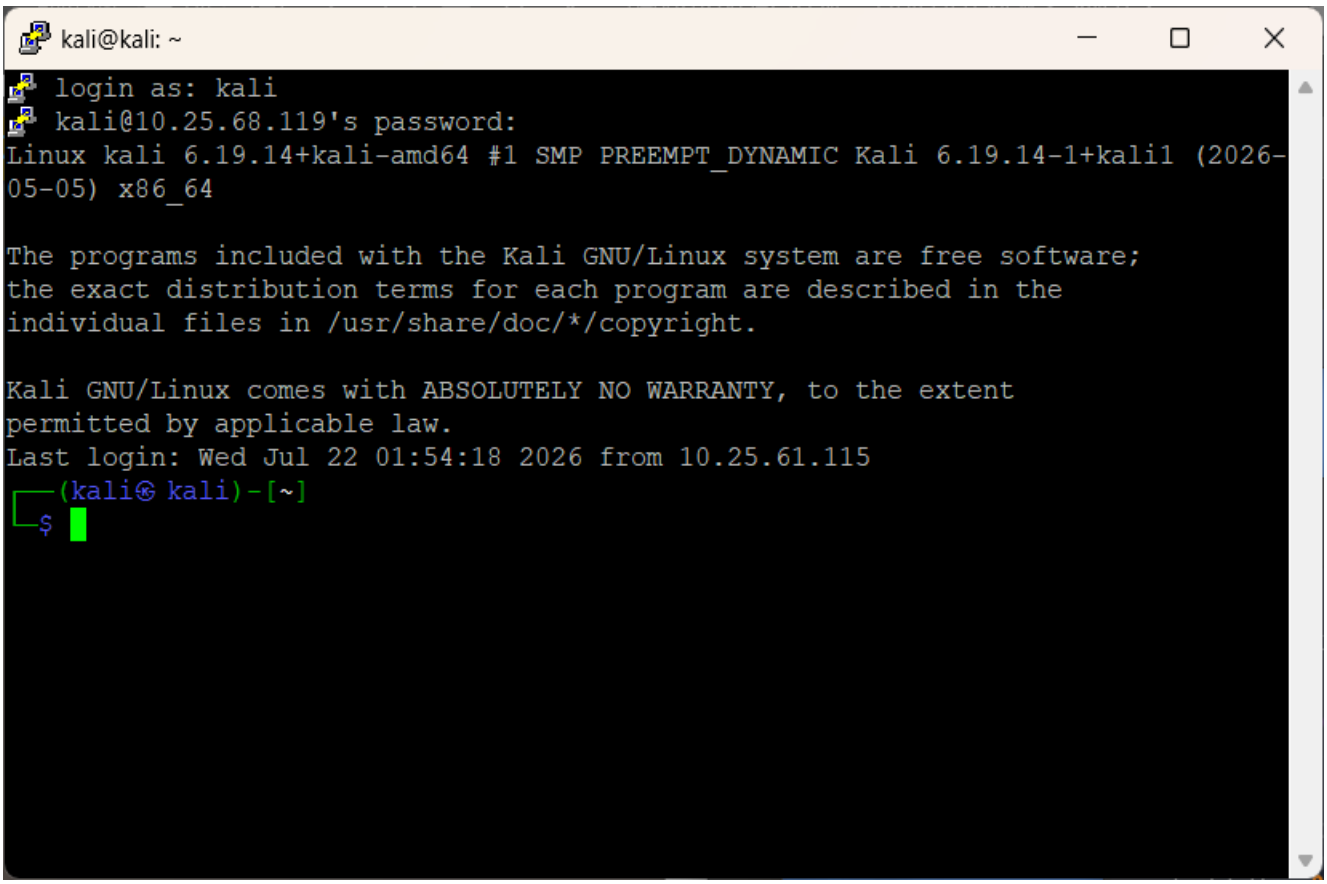
1. Screenshots & Steps of:
    - Your PC IP Address & Linux VM IP address



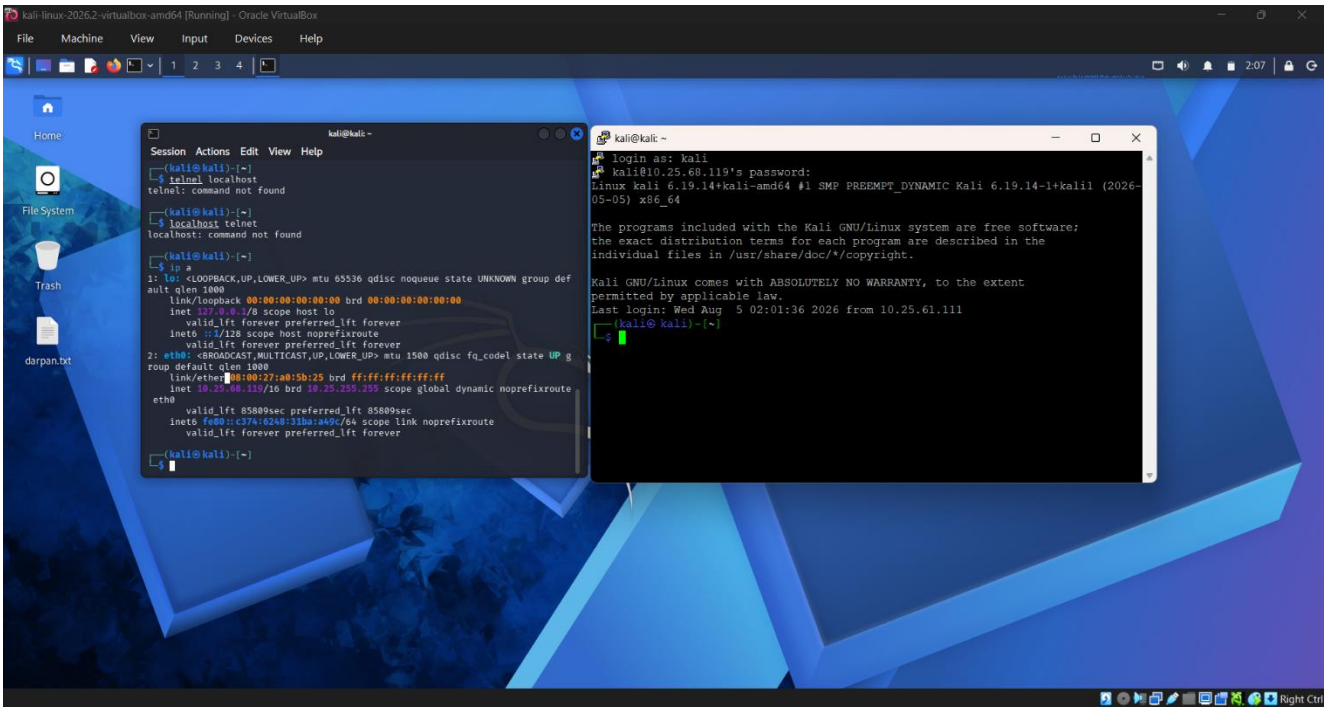
Date: 23/07/2026



- Successful SSH login using PuTTY

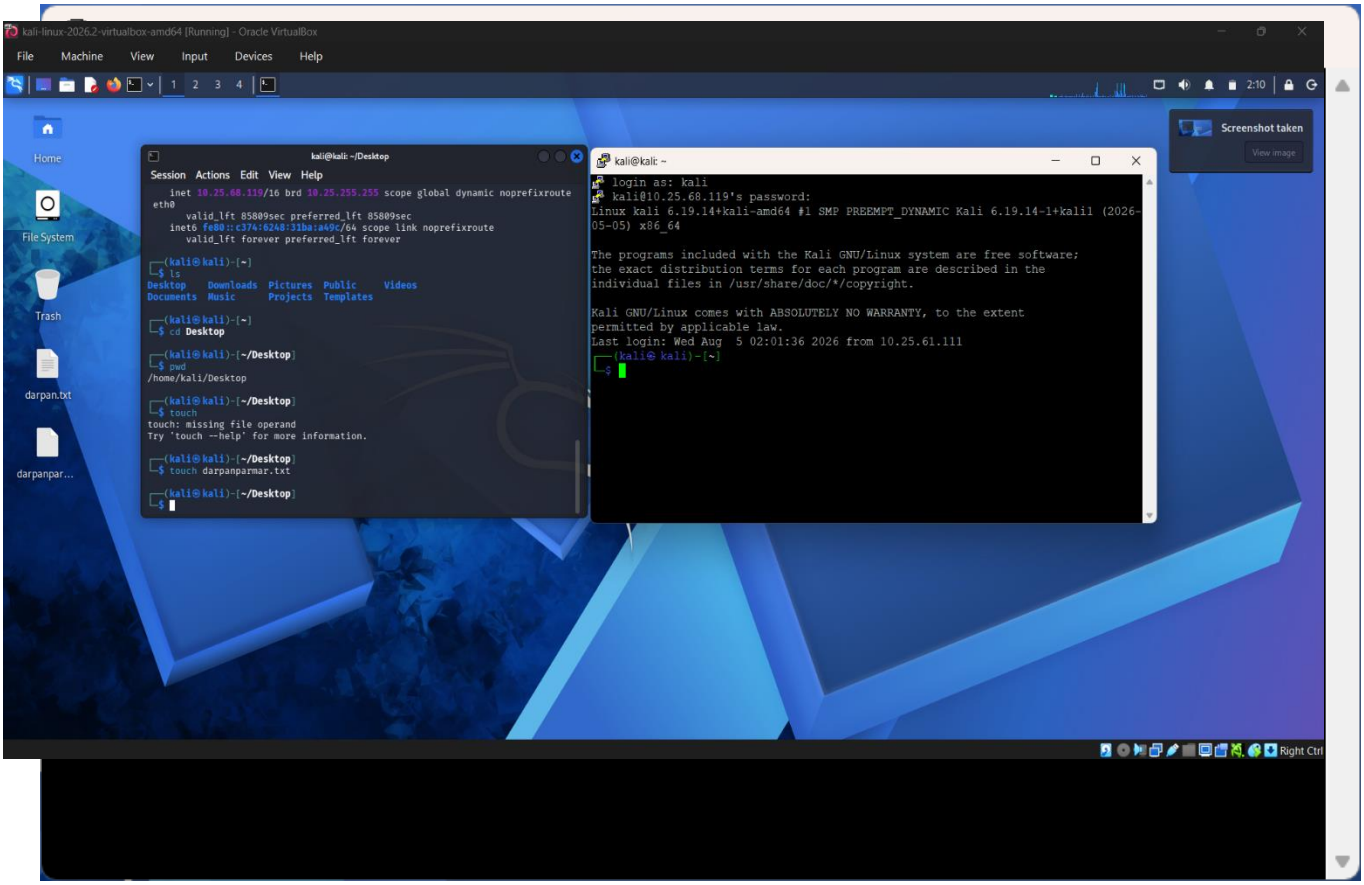


Date: 23/07/2026

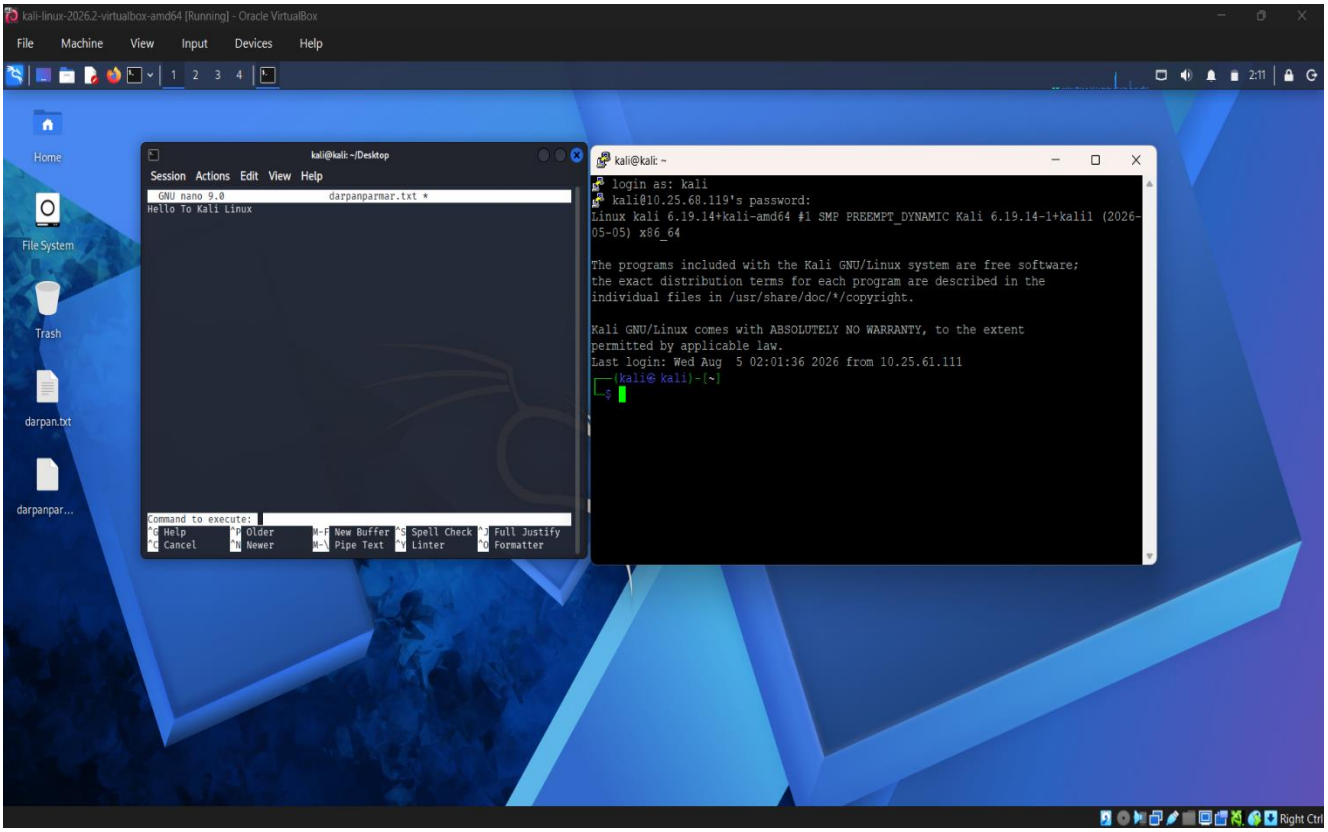


- Successful Telnet login using PuTTY

- Execution of Linux commands

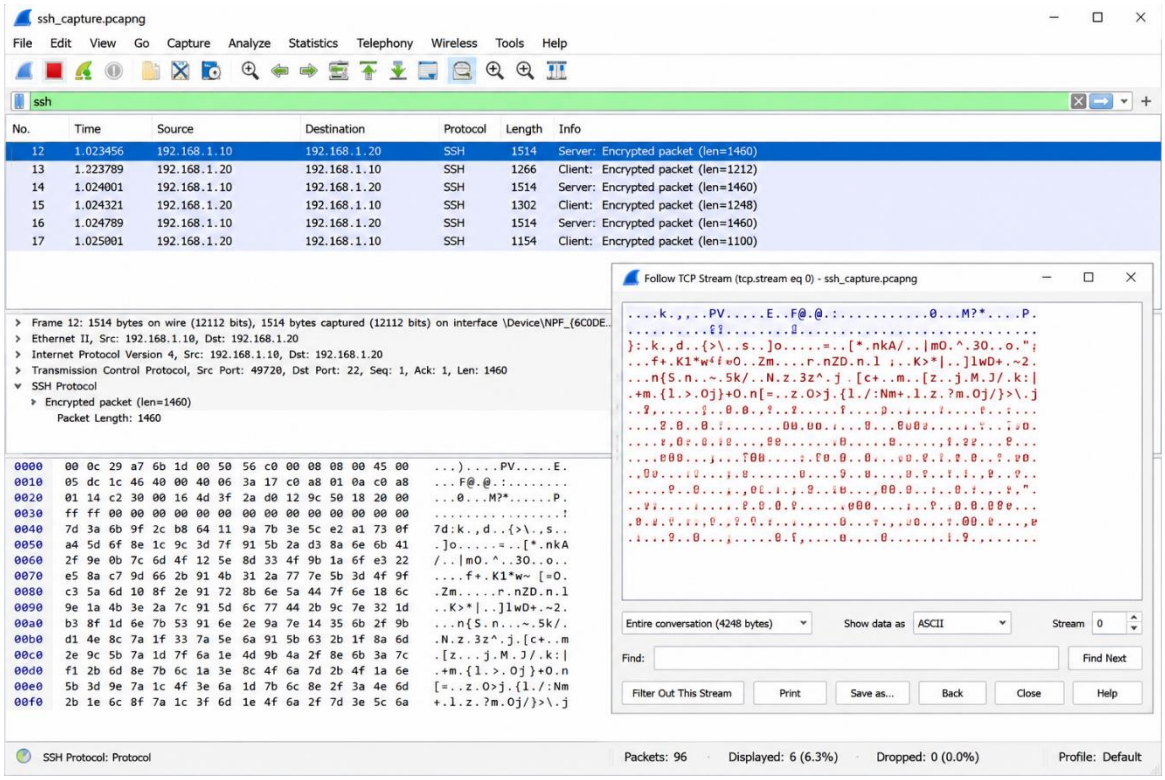


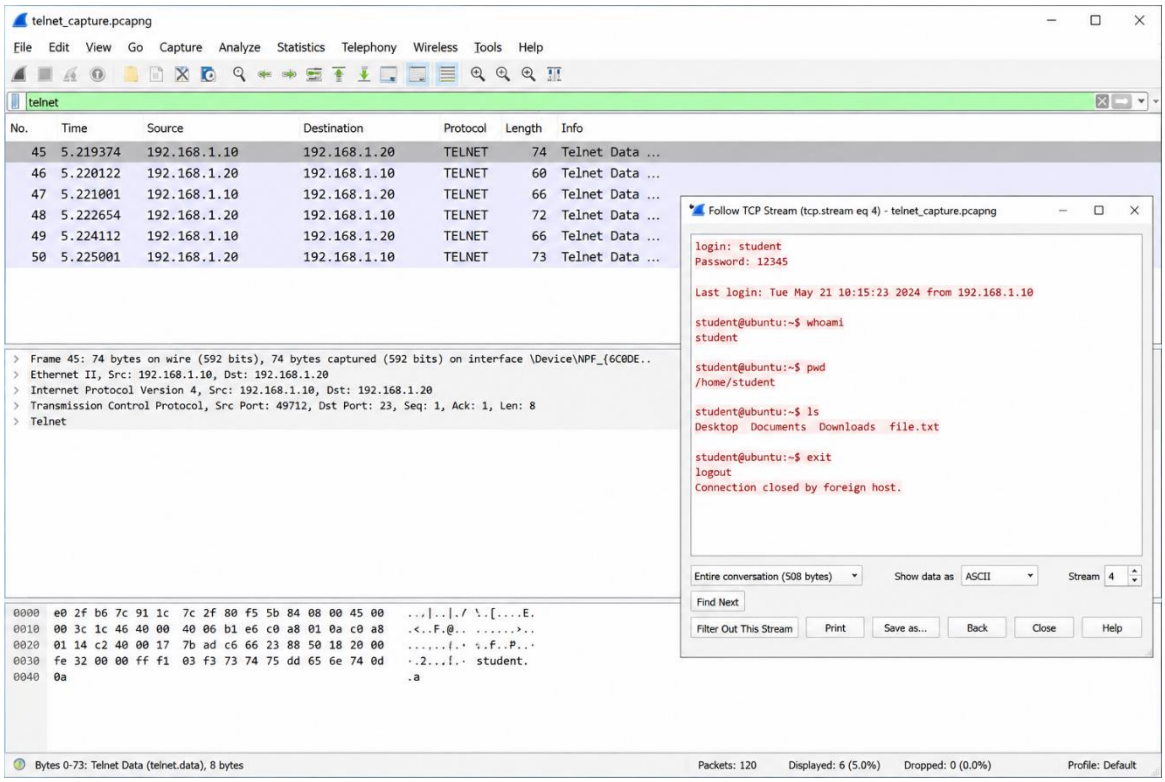
Date: 23/07/2026



Date: 23/07/2026

- Capture network packets to analyze security differences between both protocols. (plain text vs. encryption).





Date: 23/07/2026

Packet Analysis (Wireshark Observation)

Telnet

- Username and password are visible in plain text.
- Commands typed by the user can be read easily.
- Data packets are not encrypted.

SSH

- Username, password, and commands are encrypted.
- Packet contents appear as unreadable encrypted data.
- Sensitive information cannot be viewed directly.

Conclusion: SSH provides confidentiality, integrity, and secure authentication, whereas Telnet sends all information in plain text and is therefore insecure.

2. Assignment Questions
   - Explain the difference between Telnet and SSH.

Telnet	SSH (Secure Shell)
• Uses plain text communication.	• Uses encrypted communication.
• Data, including username and password, is sent without encryption.	• Username, password, and all data are encrypted.
• Default Port: 23	• Default Port: 22
• Less secure and rarely used today.	• Highly secure and widely used.
• Suitable only for trusted/private networks.	• Suitable for both public and private networks.

   - Why is SSH considered more secure?

     SSH is considered more secure because it encrypts all communication between the client and the remote server.

     Reasons:
     - Encrypts usernames and passwords.
     - Protects transmitted data from attackers.
     - Prevents packet sniffing and eavesdropping.
     - Ensures data integrity (data cannot be modified during transmission).

- Supports public key authentication for stronger security.
- Protects against Man-in-the-Middle (MITM) attacks.

- List the Linux commands executed during the practical and write their purpose.

Command	Purpose
pwd	Displays the current working directory.
ls	Lists files and directories.
ls -l	Displays detailed information about files.
cd	Changes the current directory.
mkdir test	Creates a new directory named **test**.
touch file1.txt	Creates an empty file.
cat file1.txt	Displays the contents of a file.
echo "Hello" > file1.txt	Writes text into a file.
cp file1.txt backup.txt	Copies a file.
mv backup.txt newfile.txt	Renames or moves a file.
rm newfile.txt	Deletes a file.
rmdir test	Removes an empty directory.
whoami	Displays the currently logged-in user.
hostname	Displays the system hostname.
ip addr	Shows IP address information.
ifconfig *(if available)*	Displays network configuration.
clear	Clears the terminal screen.
exit	Logs out from the SSH/Telnet session.

- Explain the role of PuTTY in remote administration.

PuTTY is a free SSH and Telnet client used to remotely access and manage Linux or Unix servers from a Windows computer.

Roles of PuTTY:

- Establishes remote SSH and Telnet connections.
- Allows administrators to execute Linux commands remotely.
- Enables server configuration and maintenance.
- Supports secure file transfer using SCP and SFTP (with related tools).
- Helps monitor and troubleshoot remote systems.

Date: 23/07/2026

- Mention three real-world applications of SSH.

1. Remote Server Administration

System administrators securely log in to Linux servers to manage files, users, and services.

2. Secure File Transfer

SSH is used with SCP or SFTP to securely transfer files between computers.

3. Cloud Server Management

Developers connect to cloud servers (AWS EC2, Azure Virtual Machines, Google Cloud VMs, DigitalOcean, etc.) using SSH for deployment and maintenance.